P vs NP as an Entropic No-Go Theorem: Algorithmic Positivity, Kolmogorov Obstruction, and the Impossibility of Witness Compression

Lukas Geiger[1,*]

[1] *Independent Researcher, Bernau im Schwarzwald, Germany*[†]

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We propose a reformulation of the P vs NP problem as an *entropic No-Go theorem*: P $\neq$ NP is equivalent to the structural impossibility of compressing NP-witnesses to algorithmic entropy below their intrinsic Kolmogorov complexity. This paper does not prove P $\neq$ NP; rather, it provides an information-theoretic reformulation that identifies the precise structure of what must be shown and why classical barrier techniques are not obstructions to this approach. We introduce the concept of *algorithmic positivity* – an NP language is algorithmically positive if it admits an entropically convex, polynomially scaling witness-producing function – and characterize the *Entropic Separation Conjecture* (no NP-complete language is algorithmically positive) as *equivalent* to P $\neq$ NP, establishing ESC as a reformulation of the P vs NP problem in information-theoretic language rather than an independent result. We conduct a systematic barrier analysis, arguing that arguments based on unbounded Kolmogorov complexity K are not naturally subject to the three classical barriers (relativization, natural proofs, algebrization), while noting that the barrier status for the resource-bounded variant K^t used in our core results remains to be fully established. The key structural insight is the *witness entropy gap*: P = NP would imply $K^t(w \mid x) = O(\log |x|)$ for all NP-witnesses, while P $\neq$ NP implies the existence of instance families with $K^t(w \mid x) \geq |w| - O(\log n)$ for every polynomial time bound t. We identify the precise remaining challenge: converting this characterization into a proof requires a *uniformity bridge* – a new connection between algorithmic information theory and uniform complexity theory.

I. INTRODUCTION

A. The problem

The P vs NP question asks whether every problem whose solutions can be *verified* in polynomial time can also be *solved* in polynomial time. Formally:

Definition I.1. P is the class of languages decidable in polynomial time. NP is the class of languages for which membership has a polynomial-time verifiable witness [11]. P = NP iff every language in NP is also in P. By the Cook-Levin theorem [20], this is equivalent to the satisfiability problem (SAT) being solvable in polynomial time.

Despite decades of effort, the problem remains open. This paper argues that the difficulty is not technical but *conceptual*: previous approaches attempt to prove lower bounds via *constructive* methods (exhibiting specific hard instances, explicit circuit separations, or algebraic obstructions), which are systematically blocked by three barriers. We propose an alternative: prove P $\neq$ NP via a *structural entropy argument* that identifies the impossibility of witness compression as the fundamental obstruction.

B. The three barriers

Three meta-theorems limit the scope of known proof techniques:

1. Relativization [1]. There exist oracles A, B with $\mathrm{P}^A = \mathrm{NP}^A$ and $\mathrm{P}^B \neq \mathrm{NP}^B$. Therefore, any proof of P $\neq$ NP must use techniques that do not relativize – it must exploit the internal structure of computation, not just input-output behavior.

2. Natural proofs [2]. If strong pseudorandom generators exist (a widely believed assumption), then no "natural" proof method – one that is both *constructive* (recognizes hard functions efficiently) and *large* (applies to a dense set of functions) – can prove superpolynomial circuit lower bounds against P/poly.

3. Algebrization [3]. Even non-relativizing techniques based on arithmetization (as in IP = PSPACE) cannot separate P from NP, because there exist algebraic extensions of oracles in which both P = NP and P $\neq$ NP hold.

C. The positivity alternative

We observe that major breakthroughs across mathematics share a common structure: they succeed not by constructing the desired object, but by showing its non-existence would violate a *positivity condition*:

* https://orcid.org/0009-0005-7296-1534

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Problem	Positive structure	Status
Weil conjectures	Frobenius eigenvalues	Proven
BSD (rank ≤ 1)	Néron-Tate height	Proven
Hodge conjecture	Hodge-Riemann form	Open
RH	Spectral zeta flow	[17]
P vs NP	**Kolmogorov complexity**	**This paper**
Broader program	Free energy framework	[16]

The key insight: *Kolmogorov complexity is the natural positivity structure for computational complexity.* It is non-negative ($K(x) \geq 0$), machine-invariant (up to additive constants), and fundamentally non-constructive (not computable) – precisely the properties needed to bypass the three barriers.

This paper is self-contained; all main results depend only on definitions and proofs within this document. Connections to the broader FST programme are cited for context but are not logically required.

II. SYSTEMATIC BARRIER ANALYSIS

A. Why constructive methods fail

All three barriers share a common cause: they block proofs that treat computational problems as *black boxes* or that rely on *statistical* properties of Boolean functions.

- **Relativization** blocks proofs that treat the computation as input-output behavior (oracle access), ignoring internal structure.
- **Natural proofs** block proofs that identify hard functions via efficiently testable, statistically dense properties.
- **Algebrization** blocks proofs that use only the polynomial structure of arithmetized computations.

The common thread: all three barriers apply to *model-dependent* arguments – arguments that work within a specific computational model (oracle machines, circuits, algebraic extensions) rather than targeting the *intrinsic information content* of the computation.

B. The specification for a valid proof

From the barrier analysis, we derive necessary conditions for any proof of P $\neq$ NP:

(S1) Non-relativizing: The argument must exploit internal computational structure, not just oracle behavior.

(S2) Non-natural: The argument must not define a "largeness" property that is efficiently computable. It must target individual objects or non-constructive properties.

(S3) Non-algebrizing: The argument must go beyond polynomial identity testing and arithmetization.

(S4) Model-independent: The argument must not depend on a specific computational model (circuits, Turing machines, etc.) but on an invariant structural property.

(S5) Robust: The argument must not break when moving from restricted models (monotone circuits, AC^0) to general computation.

We now show that algorithmic entropy satisfies all five specifications.

III. ALGORITHMIC ENTROPY AS POSITIVITY STRUCTURE

A. Kolmogorov complexity

Let U be a fixed universal prefix-free Turing machine. The *Kolmogorov complexity* (algorithmic entropy) of a finite binary string x is:

$$K(x) := \min\{|p| : U(p) = x\}\,. \tag{1}$$

The *conditional* Kolmogorov complexity of x given y is:

$$K(x \mid y) := \min\{|p| : U(p, y) = x\}\,. \tag{2}$$

Theorem III.1 (Invariance [12, 13, 22])**.** *For any two universal prefix-free machines U, U':*

$$|K_U(x) - K_{U'}(x)| \leq c_{U,U'}\,, \tag{3}$$

where $c_{U,U'}$ is a constant independent of x.

Theorem III.2 (Incomputability)**.** *$K(x)$ is not computable. More precisely: for every total computable function f, there exists x with $K(x) > f(x)$.*

Theorem III.3 (Incompressibility)**.** *For every n, at least $2^n - 2^{n-c+1} + 1$ strings of length n satisfy $K(x) \geq n - c$.*

Theorem III.4 (Symmetry of Information [4])**.** *For all strings x, y:*

$$K(x, y) = K(x) + K(y \mid x, K(x)) + O(\log(K(x, y)))\,. \tag{4}$$

This implies that conditional complexity is well-behaved: the information in the pair (x, y) decomposes additively (up to logarithmic terms) into the information in x plus the additional information in y given x.

B. Why Kolmogorov complexity bypasses the barriers

Claim III.5 (Barrier immunity). Arguments based on Kolmogorov complexity plausibly satisfy specifications (S1)–(S5). We present heuristic arguments for each; a fully rigorous treatment is an open research question (see Limitations, Section XI).

(S1) Non-relativizing. $K(x)$ is defined without reference to any oracle – it is an absolute, fixed quantity for each string x. The witness entropy gap (Theorem IV.3) uses $K^t(w \mid x)$ (resource-bounded conditional complexity). We note an important subtlety: the invariance theorem for resource-bounded K^t is weaker than for unbounded K – changing the universal machine can introduce a *polynomial* slowdown factor rather than just an additive constant [4]. However, the argument remains non-relativizing because K^t is defined over a fixed computational model without oracle access. A relativizing proof would need to hold for all oracles A, but our argument targets oracle-independent quantities: the gap between $O(\log n)$ and $\Omega(|w|)$ is superpolynomial and thus robust under polynomial slowdown factors.

(S2) Non-natural. $K(x)$ is not computable (Theorem III.2), so no efficient procedure can test whether a specific function has "high Kolmogorov complexity." This directly violates the "constructiveness" requirement of natural proofs.

(S3) Non-algebrizing. Algebrization applies to proof techniques that use only the polynomial structure of arithmetized computations. Arguments based on K^t operate over the full power of Turing machine computation, not polynomial identity testing. While it remains a research question whether K^t-based arguments are formally non-algebrizing in the technical sense of [3], the entropy framework is not naturally expressible as an algebraic identity, suggesting it escapes this barrier.

(S4) Model-independent. The invariance theorem ensures that $K(x)$ is the same (up to $O(1)$) for all universal machines – Turing machines, circuits, λ-calculus, etc.

(S5) Robust. $K(x)$ does not depend on the computational model used to define it.

Remark III.6 (Barrier immunity: unbounded K vs. resource-bounded K^t). The barrier immunity claimed above applies to unbounded Kolmogorov complexity K, which is an absolute, uncomputable quantity invariant up to $O(1)$ under change of universal machine. However, the core technical results of this paper (Theorem IV.3, the witness entropy gap) necessarily use *resource-bounded* K^t, which is computable and whose invariance is weaker (polynomial slowdown rather than additive constant). The barrier status of K^t-based arguments is therefore less clear-cut:

- **Relativization:** $K^t(w \mid x)$ is defined without oracles, so our arguments are non-relativizing in the sense that they do not hold "for all oracles." However, $K^{t,A}(w \mid x)$ (with oracle access) can differ substantially from $K^t(w \mid x)$, and a full barrier-immunity proof for the resource-bounded case remains open.
- **Natural proofs:** K^t is computable, so unlike K, it does not automatically escape the constructiveness requirement. The argument that our approach avoids natural proofs rests on the fact that we do not use K^t as an efficiently testable "largeness" property of Boolean functions, but this distinction deserves further formalization.
- **Algebrization:** No formal proof exists that K^t-based arguments are non-algebrizing in the technical sense of [3]. S3 (non-algebrization for K^t) remains an open question; we make no claim that our framework provably evades this barrier.

In summary: the barrier bypass is rigorous for unbounded K but remains heuristic for resource-bounded K^t. Since the witness entropy gap requires K^t, establishing formal barrier immunity for the resource-bounded case is an important open problem.

IV. THE WITNESS ENTROPY GAP

A. NP witnesses as compressed information

For an NP language L, every yes-instance $x \in L$ has a polynomial-length witness w verifiable in polynomial time:

$$x \in L \iff \exists w,\ |w| \le \mathrm{poly}(|x|),\ V(x,w) = 1\,. \tag{5}$$

The conditional complexity of the witness given the instance measures how much *additional* information the witness contains beyond the instance:

$$K(w \mid x) = \text{“extra information needed to produce } w \text{ from } x\text{”}\,. \tag{6}$$

B. What P = NP would imply

Proposition IV.1 (Witness compression under P = NP)**.** *If* P = NP*, then for every NP language L and every $x \in L$, a valid witness w can be produced from x by a polynomial-time algorithm A:*

$$w = A(x)\,. \tag{7}$$

Therefore, for some polynomial q depending on the running time of A:

$$K^{q(|x|)}(w \mid x) \leq K(A) + O(\log |x|) = O(\log |x|)\,, \quad (8)$$

since the algorithm A is a fixed finite program running in polynomial time, and $O(\log |x|)$ bits encode the input length.

Proof. If P = NP, the search version of SAT is solvable in polynomial time (by self-reducibility). Given a satisfiable formula φ, the algorithm A finds a satisfying assignment bit-by-bit in polynomial time. The program "run A on input x" has length $|A| + O(\log |x|)$, where $|A|$ is a constant. □

C. The entropy obstruction

Remark IV.2 (Unbounded vs. resource-bounded complexity – why K^t is essential). The distinction between $K(w \mid x)$ and $K^t(w \mid x)$ is crucial and must be understood before the main theorem. For any decidable language L and any (x, w) with $V(x, w) = 1$, the brute-force verifier-search has $K(w \mid x) \leq O(1)$ (the search program has constant length – it simply enumerates all candidates and checks each). The information-theoretic content of NP-hardness lies entirely in the *time* required to extract the witness, not in the *description length*. This is why K^t with polynomial t is the correct measure: it captures the computational cost of witness extraction, which is the essence of the P vs NP question.

Theorem IV.3 (Existence of high-entropy witnesses – resource-bounded, conditional)**.** *If* P $\neq$ NP*, then for every NP-complete language L, there exist infinitely many yes-instances $x \in L$ such that for every polynomial q,* ***every*** *valid witness w for x satisfies (for the unique satisfying assignment, the bound is sharpest; for instances with multiple witnesses, it applies to every witness simultaneously):*

$$K^{q(|x|)}(w \mid x) \geq |w| - O(\log |x|)\,. \quad (9)$$

That is, no polynomial-time program given x can produce w. (The "infinitely many" follows because finitely many exceptions can be hardcoded as a constant-size lookup table, leaving a polynomial-time algorithm for all remaining instances.)

Equivalently (by contraposition): if there exist a polynomial q and a constant c such that for every NP-complete language L and every instance $x \in L$, some valid witness w satisfies $K^{q(|x|)}(w \mid x) \leq c$, then P = NP*.*

Note on quantifier order: the theorem asserts that a single *infinite set of hard instances resists all polynomials q simultaneously. This follows because $K^{q_1}(w \mid x) \leq K^{q_2}(w \mid x)$ whenever $q_1 \geq q_2$ (more time can only decrease description length). Therefore, if an instance x has a witness w with $K^{q_0(|x|)}(w \mid x) \geq |w| - O(\log |x|)$ for some polynomial q_0, the same bound holds for all $q \leq q_0$. The contrapositive guarantees that for every fixed q, hard instances exist; the monotonicity of K^t in t ensures that instances that are hard for large time bounds are automatically hard for all smaller ones, so the infinite set can be chosen independently of q.*

Proof. We prove the contrapositive. Suppose for some NP-complete language L, some polynomial q, and some constant c, every yes-instance $x \in L$ has a witness w with $K^{q(|x|)}(w \mid x) \leq c$. Then for every yes-instance $x \in L$, there exists a program p_x of length $\leq c$ that, given x, produces a valid witness w in time $q(|x|)$. (Note: p_x may depend on x, but there are at most $2^{c+1} - 1$ programs of length $\leq c$.) This gives a polynomial-time search algorithm for L: on input x, enumerate all programs of length $\leq c$, run each with input x for at most $q(|x|)$ steps, and verify each output. Since L is NP-complete, every NP search problem reduces to the search version of L via polynomial-time reductions. A polynomial-time search algorithm for L therefore yields polynomial-time search algorithms for all NP search problems, implying FP = FNP (where FP and FNP are the function-problem analogues of P and NP) and hence P = NP.

Existence of infinitely many high-entropy instances (given P $\neq$ NP*):* The formal proof is the contrapositive above. The *existence* of high-entropy instances is guaranteed non-constructively: if every SAT instance had a low-entropy witness ($K^{q(n)}(w \mid x) \leq c$ for some fixed q, c), we would have P = NP. Under P $\neq$ NP, such instances must exist. Moreover, there must be *infinitely many*: if only finitely many hard instances existed, they could be hardcoded into a constant-size lookup table, leaving a polynomial-time algorithm for all remaining instances – contradicting P $\neq$ NP. Constructing the hard instances explicitly is non-trivial – a naive approach via Cook-Levin encoding of "verify $v = r$" for a random string r fails because r is recoverable from the formula structure in polynomial time, giving $K^{\text{poly}}(r \mid \varphi_r) = O(1)$. The extension to any NP-complete language L follows from NP-completeness: since SAT reduces to L in polynomial time, a polynomial-time witness-finding algorithm for L composed with the reduction yields a polynomial-time witness-finding algorithm for SAT (one reduces a SAT instance to an L-instance, finds a witness for L, and uses the witness to reconstruct a satisfying assignment for the original SAT instance), assuming the reduction is injective (which holds for standard Cook-Levin reductions).

Why unbounded K fails: For any SAT instance x with a satisfying assignment w, the brute-force program "enumerate all assignments, test each, output the first satisfying one" has length $O(1)$ and finds w from x in time $O(2^{|x|})$. Thus $K(w \mid x) \leq O(1)$ – the unbounded conditional complexity is trivially small. This is why the resource bound is essential: the information-theoretic content of NP-hardness lies in the *time* required to extract the witness, not in the description length. □

D. The gap

Combining Proposition IV.1 and Theorem IV.3:

Corollary IV.4 (Witness Entropy Gap – the core contradiction structure)**.** *Consider the infinitely many hard instances whose existence is guaranteed by Theorem IV.3 (under* $\mathrm{P} \neq \mathrm{NP}$*). For unique-witness instances, the contradiction structure is sharpest:*

- $\mathrm{P} = \mathrm{NP} \Rightarrow K^{q(|x|)}(w \mid x) \leq O(\log |x|)$ *for some polynomial* q *(Proposition IV.1), since any algorithm* A *must output the unique witness* $w = A(x)$*.*
- $\mathrm{P} \neq \mathrm{NP} \Rightarrow K^{q(|x|)}(w \mid x) \geq |w| - O(\log |x|)$ *for all polynomials* q *(Theorem IV.3).*

This is not a proof of $\mathrm{P} \neq \mathrm{NP}$*, but a sharp characterization:* $\mathrm{P} = \mathrm{NP}$ *if and only if all NP-witnesses are polynomially compressible. The entropy gap* $\Omega(|w|)$ *vs.* $O(\log n)$ *quantifies exactly what separating* P *from* NP *requires.*

The gap established here is conditional on the instance having a unique (or low-multiplicity) satisfying assignment; for instances with many witnesses, the algorithm may select a low-complexity witness, circumventing the entropy barrier.

Remark IV.5 (Unique vs. multiple witnesses). Theorem IV.3 guarantees instances where *every* valid witness has high conditional complexity. For these instances, the contradiction structure holds regardless of how many witnesses exist: any algorithm A must output *some* valid witness, and *all* of them satisfy the high-entropy bound (9).

For unique-witness instances, the structure is especially clean: A has no choice over which witness to produce. In general, however, the high-entropy instances guaranteed by Theorem IV.3 exist non-constructively (via contraposition), and it remains open whether they can be constructed explicitly – see Section VIII.

V. ALGORITHMIC POSITIVITY

We now formalize the No-Go framework.

A. Definition

Definition V.1 (Algorithmic Positivity)**.** An NP language L (with verifier V) is *algorithmically positive* if there exists a *witness-producing* function $W : \{0,1\}^* \to \{0,1\}^*$ satisfying (AP1) and (AP2):

(AP1) Polynomial scaling. W is computable in time $\mathrm{poly}(|x|)$, and for every $x \in L$: $V(x, W(x)) = 1$.

(AP2) Entropic convexity. There exists a polynomial q such that for every $x \in L$:

$$K^{q(|x|)}(W(x) \mid x) \leq O(\log |x|)\,. \tag{10}$$

The witness production is "entropically cheap": the witness $W(x)$ adds at most logarithmic resource-bounded information beyond the input. (Note: since W is a fixed polynomial-time program, the polynomial q is determined by W and does not depend on the instance x.)

Remark V.2. Note that (AP2) targets the *witness* $W(x)$, not just the decision bit. A single decision bit $D(x) \in \{0,1\}$ trivially satisfies $K(D(x) \mid x) \leq 1$, so entropic convexity would be vacuous for decision functions. The meaningful measure is the resource-bounded complexity of producing a *witness* – this is where the information-theoretic content of NP-hardness resides.

Remark V.3. Every P-decidable NP language is algorithmically positive: if $L \in \mathrm{P} \cap \mathrm{NP}$, then the polynomial-time search algorithm (obtained via self-reducibility for NP-complete problems, or directly for natural NP problems) produces a witness $W(x)$ with $K^{q(|x|)}(W(x) \mid x) \leq |W_{\mathrm{prog}}| + O(\log |x|) = O(\log |x|)$.

Remark V.4 (Heuristic properties of P-decidable NP languages). For NP languages in P, the witness-producing function W additionally satisfies two heuristic properties that capture the "structural simplicity" of polynomial-time witness extraction:

(H1) Compositional stability. For structured instances under a well-defined composition operator $\circ$, the witness for the composed input is typically determined by the component witnesses with at most logarithmic overhead: $K^{\mathrm{poly}}(W(x_1 \circ x_2) \mid W(x_1), W(x_2)) \leq O(\log(|x_1| + |x_2|))$.

(H2) Monotone stability. Small perturbations of the input produce witnesses of bounded additional complexity: $d_H(x, x') \leq 1 \implies K(W(x) \mid W(x'), x) \leq O(\mathrm{poly}(\log |x|))$.

These properties are motivated by the structure of NP-complete problems but are not used in the formal No-Go argument (Theorem VI.1). They serve as additional heuristic evidence for the Entropic Separation Conjecture.

B. The Entropic Separation Conjecture

Conjecture V.5 (Entropic Separation – ESC)**.** *No NP-complete language is algorithmically positive. That is: for every NP-complete language* L *and every polynomial-time witness-producing function* W*, there exist infinitely many yes-instances* $x \in L$ *such that* W *fails to produce a valid witness for* x *(i.e.,* $V(x, W(x)) = 0$*), or* $K^{q(|x|)}(W(x) \mid x) \geq \omega(\log |x|)$ *for all polynomials* q*.*

An equivalent formulation in terms of truth-table complexity (see Theorem VII.4): for some NP-complete language L*, for every polynomial* q*, infinitely many* n *satisfy* $K^{q(2^n)}(L_n) \geq n$*. (Both formulations are equivalent to* $\mathrm{P} \neq \mathrm{NP}$*; their mutual equivalence follows transitively.)*

Proposition V.6. *The Entropic Separation Conjecture V.5 implies* $\mathrm{P} \neq \mathrm{NP}$*.*

Proof. If P = NP, then for every NP-complete language L, the search version is solvable in polynomial time (for SAT, by self-reducibility; for general NP-complete L, by reducing L to SAT, applying the SAT search algorithm, and reconstructing the witness). This gives a polynomial-time witness-producing function W satisfying (AP1). Since W is a fixed program running in polynomial time, $K^{q(|x|)}(W(x) \mid x) \leq |W_{\text{prog}}| + O(\log |x|) = O(\log |x|)$, satisfying (AP2). Therefore, NP-complete languages would be algorithmically positive, contradicting the conjecture. □

Remark V.7 (ESC is equivalent to P ≠ NP). The converse of Proposition V.6 also holds: P ≠ NP implies ESC. Indeed, if P = NP, then every NP-complete language L has a polynomial-time algorithm, giving $K^{\text{poly}(2^n)}(L_n) = O(\log n)$ by Lemma VII.3, which violates the ESC condition $K^{q(2^n)}(L_n) \geq n$. Thus ESC ⇔ P ≠ NP: the Entropic Separation Conjecture is a *reformulation* of the P vs NP problem in information-theoretic language, not an independent strengthening.

C. Why NP-complete problems resist algorithmic positivity

The failure of each condition for NP-complete problems:

(AP2) fails: For NP-complete problems, producing a witness w for $x \in L$ requires extracting information that is algorithmically random relative to the instance. The resource-bounded conditional complexity $K^t(w \mid x)$ can be $\Omega(|w|)$ (Theorem IV.3, conditional on P ≠ NP). No polynomial-time witness-producing function W can achieve $K^{\text{poly}}(W(x) \mid x) \leq O(\log |x|)$ for all instances.

As additional heuristic evidence (not part of the formal argument), the properties (H1)–(H2) from Remark V.4 also fail for NP-complete problems:

(H1) fails: SAT is not compositionally decomposable in the required sense. The satisfiability of a conjunction $\varphi_1 \wedge \varphi_2$ is not determined by the satisfiability of φ_1 and φ_2 individually (they share variables). This non-decomposability is precisely the source of NP-hardness.

(H2) fails: For random k-SAT instances near the satisfiability threshold, SAT exhibits extreme sensitivity in its search version: flipping a single clause can restructure the entire witness space. Specifically, near the phase transition (clause-to-variable ratio $\alpha \approx \alpha_c(k)$), a single clause addition can fragment the solution cluster into exponentially many components, producing witnesses of unbounded additional complexity [26, 27]. For structured or under-constrained instances, this sensitivity may be less pronounced; the heuristic argument is strongest in the random near-threshold regime.

VI. THE NO-GO THEOREM

Theorem VI.1 (Entropic No-Go – Reformulation)**.** *The following are equivalent:*

(i) P ≠ NP.

(ii) The Entropic Separation Conjecture V.5 holds.

(iii) No NP-complete language is decidable in polynomial time. That is: for every NP-complete language L and every polynomial-time algorithm A, there exist instances x where $A(x) \neq L(x)$.

The equivalence (i) ⇔ (ii) shows that P ≠ NP *is equivalent to an entropy obstruction: polynomial-time algorithms cannot compress NP-witnesses below their intrinsic Kolmogorov complexity.*

Proof. $(ii) \Rightarrow (i)$: Shown in Proposition V.6. Contrapositive: if P = NP, then every NP-complete language has a polynomial-time witness-producing function (via the SAT search algorithm and polynomial-time reductions), satisfying both (AP1) and (AP2). This makes NP-complete languages algorithmically positive, contradicting ESC.

$(i) \Rightarrow (ii)$: Contrapositive: suppose ESC fails, i.e., some NP-complete language L is algorithmically positive. Then L has a polynomial-time witness-producing function W with $K^{\text{poly}}(W(x) \mid x) \leq O(\log |x|)$. In particular, W is a polynomial-time search algorithm for L. Since L is NP-complete, this implies P = NP.

$(i) \Leftrightarrow (iii)$: (iii) states that for every NP-complete language L, $L \notin$ P. This is equivalent to P ≠ NP by the existence of NP-complete languages (if any NP-complete L were in P, then P = NP; conversely, P = NP would place every NP-complete language in P).

The content of the theorem is not a new separation result, but the *reformulation*: P ≠ NP is equivalent to an entropic incompressibility condition on NP-witnesses. This reformulation identifies the precise structure (witness entropy) and the precise gap (uniformity bridge) that any proof must address. □

A. Exclusion of alternative scenarios

Proposition VI.2 (Excluded scenarios)**.** *The following scenarios are incompatible with the conjunction of entropic positivity and Kolmogorov obstruction:*

(X1) Polynomial algorithm with exponential entropy production. *An algorithm that "invents" new information during computation. Excluded because for any deterministic algorithm A running in time $t_A(|x|)$, the resource-bounded conditional complexity satisfies $K^{t_A(|x|)}(A(x) \mid x) \leq |A| + O(\log |x|)$: the program "run A on input x" has constant description length. In particular, the conditional algorithmic information of $A(x)$ given x*

is at most $O(1)$ bits at the time scale of A's own computation – meaning A cannot "create" information that was not already implicit in the input. (For randomized algorithms, the random bits r provide additional input, so $K(A(x,r) \mid x)$ can be high. However, the P vs NP question concerns deterministic computation; moreover, under widely believed derandomization assumptions, BPP = P *[15], so randomness does not provide additional power.)*

(X2) Compositional collapse of NP. *All SAT instances decompose into independently solvable parts. Excluded by the NP-completeness of SAT: if SAT decomposed, then all NP problems would decompose (via reduction), but UNIQUE-SAT instances with random solutions do not decompose. (The relevance of UNIQUE-SAT is supported by the Valiant-Vazirani theorem [21]: SAT reduces to UNIQUE-SAT under randomized reductions.)*

(X3) Smooth decision landscape. *All SAT instances have monotonically stable decision boundaries. Excluded by phase-transition phenomena in random SAT: near the satisfiability threshold, the decision changes discontinuously under small clause additions.*

VII. RESOURCE-BOUNDED SLICE ENTROPY

The witness entropy gap of Section IV targets individual instances. A cleaner formulation targets the *truth table* of the language on all inputs of a given length.

A. Slices of a language

Definition VII.1 (Slice)**.** For a language $L \subseteq \{0,1\}^*$, the n-th *slice* $L_n \in \{0,1\}^{2^n}$ is the truth table of L on inputs of length n: $(L_n)_i = 1 \iff x_i \in L$, where $x_1, \ldots, x_{2^n}$ are the n-bit strings in lexicographic order.

Definition VII.2 (Resource-bounded algorithmic entropy)**.** For a time bound $t(\cdot)$:

$$K^t(x) := \min\{|p| : U(p) = x \text{ in at most } t(|x|) \text{ steps}\}\,. \tag{11}$$

B. The slice entropy lemma

Lemma VII.3 (P implies low slice entropy)**.** *If $L \in \mathrm{P}$, then there exists a polynomial q and a constant c such that for all n:*

$$K^{q(2^n)}(L_n) \le c + \lceil \log n \rceil\,. \tag{12}$$

Proof. Let A be a polynomial-time algorithm deciding L in time n^d. The program p: "read n from input; for each $x \in \{0,1\}^n$, run $A(x)$ and output the result bit" has length $|A| + O(\log n)$ (to encode n). Its runtime is $2^n \cdot n^d = \mathrm{poly}(2^n)$. Thus $K^{q(2^n)}(L_n) \le |A| + O(\log n) = c + \lceil \log n \rceil$. □

C. The No-Go theorem via slice entropy

Theorem VII.4 (Slice Entropy No-Go)**.** *If there exists an* NP*-complete language L satisfying the Entropy Hardness Hypothesis:*

(EH) *For every polynomial q, there exist infinitely many n with*

$$K^{q(2^n)}(L_n) \ge n\,, \tag{13}$$

then $\mathrm{P} \neq \mathrm{NP}$.

Proof. Immediate from Lemma VII.3: $L \in \mathrm{P}$ would give $K^{\mathrm{poly}(2^n)}(L_n) = O(\log n)$, contradicting (13) for large n. □

Remark VII.5. The Entropy Hardness Hypothesis (EH) is a concrete, well-defined conjecture about a specific NP-complete language (e.g., SAT). It asserts that the truth table of SAT on n-bit formulas cannot be compressed to fewer than n bits, even when the compression algorithm is allowed $\mathrm{poly}(2^n)$ time.

This is the *cleanest* No-Go formulation: the "easy direction" ($L \in \mathrm{P} \Rightarrow$ low entropy) is a theorem; the "hard direction" (EH for SAT) is the entire content of $\mathrm{P} \neq \mathrm{NP}$.

Note that the threshold n in EH is much smaller than the truth table length 2^n: a random string of length 2^n has $K \approx 2^n$. The threshold n is chosen because Lemma VII.3 gives $K^{\mathrm{poly}(2^n)}(L_n) = O(\log n)$ for $L \in \mathrm{P}$; any threshold $\omega(\log n)$ would suffice to separate from P, and n provides a clean, moderate lower bound that is already superpolynomial in the input length.

Remark VII.6 (Encoding sensitivity of EH). The Entropy Hardness Hypothesis depends on the encoding of SAT instances as binary strings. Different encodings yield truth tables SAT_n of different lengths and potentially different Kolmogorov complexities. The hypothesis should be understood as: there exists a *standard* polynomial-time encoding enc : Formulas $\to \{0,1\}^*$ such that $K^{q(2^n)}(\mathrm{SAT}_n^{\mathrm{enc}}) \ge n$ for infinitely many n. For resource-bounded K^t, changing the universal machine introduces a polynomial slowdown rather than just an additive constant. However, since EH quantifies over *all* polynomials q, a polynomial slowdown in the time bound does not affect the hypothesis: if $K^{q(2^n)}(\mathrm{SAT}_n) \ge n$ for all q, the same holds after re-parameterization. The *length* of the truth table (and hence the threshold n) is encoding-dependent.

Remark VII.7 (Connection to circuit lower bounds). The Entropy Hardness Hypothesis is closely related to circuit lower bounds. A Boolean function f_n computed by a circuit of size s has truth table describable in $O(s \log s)$ bits, so $K^{O(s \cdot 2^n)}(f_n) \le O(s \log s) + O(\log n)$. Thus:

- EH with threshold n implies SAT_n requires circuits of size $s \geq 2^{\Omega(n/\log n)}$, i.e., $\mathrm{NP} \not\subseteq \mathrm{SIZE}(2^{o(n)})$.
- In particular, EH implies $\mathrm{NP} \not\subseteq \mathrm{P/poly}$ (since polynomial-size circuits give $K^{\mathrm{poly}(2^n)}(f_n) \leq O(n^c \log n)$, which is $o(n)$ for large n).

This shows that the entropy framework and the circuit complexity approach are two perspectives on the same underlying separation.

Remark VII.8 (SAT slice entropy at the phase transition). The slice entropy $H_{\mathrm{slice}}(\mathrm{SAT}_n)$ was computed for random 3-SAT instances at the phase transition $\alpha = m/n \approx 4.267$ for $n = 8, 10, 12, 14, 16$. The entropy grows from $H \approx 1.5$ ($n = 8$) to $H \approx 2.3$ ($n = 14$), consistent with subexponential growth $H \sim n^{0.71}$ (polynomial fit $R^2 = 0.82$). The entropy is maximal on the satisfiable side ($\alpha \approx 2$–3) and drops to zero for $\alpha \geq 8$ (UNSAT regime), confirming the phase transition structure. Full validation of the ESC hypothesis $H_{\mathrm{slice}} \geq c \cdot 2^{n/2}$ requires $n \geq 30$, which is computationally infeasible on standard hardware. Script: `compute_sat_entropy.py`.

D. Why (EH) should be true

The truth table SAT_n has length 2^n and describes the satisfiability of all n-bit encodings of Boolean formulas. There are 2^{2^n} possible truth tables of this length, and SAT_n is a *specific* one determined by a complete mathematical definition (SAT). Nevertheless:

1. **Pseudorandom structure.** The satisfiability boundary in the space of formulas exhibits phase-transition behavior [14]: near the threshold, the pattern of satisfiable vs. unsatisfiable instances has high apparent randomness.
2. **No known compression.** Despite decades of SAT-solver research, no method compresses SAT_n to $o(2^n)$ bits in $\mathrm{poly}(2^n)$ time. The best algorithms (CDCL solvers) run in time $2^{\Theta(n)}$ in the worst case.
3. **Cryptographic evidence.** If (EH) fails, then efficient PRGs from one-way functions would not exist [15], collapsing much of modern cryptography.

VIII. THE PRECISE OBSTRUCTION

A. The uniformity gap

The main argument (Section IV) has a precise gap that we now identify.

What is proven: There *exist* NP instances whose witnesses have high conditional Kolmogorov complexity (Theorem IV.3). This is a *non-uniform* statement about individual strings.

What is needed: A *uniform* statement: for every polynomial-time algorithm A, there exist infinitely many instances x where $A(x)$ fails. This requires converting the existential entropy bound into an *adversarial* bound against all efficient algorithms simultaneously.

Definition VIII.1 (The uniformity bridge)**.** Let L be an NP-complete language with verifier V. The *uniformity bridge* is the statement: for every polynomial-time computable function $A : \{0,1\}^* \to \{0,1\}^*$, there exist infinitely many yes-instances $x \in L$ such that A fails to produce a valid witness:

$$\forall A \in \mathrm{FP},\ \exists^\infty x \in L : V(x, A(x)) = 0\,. \tag{14}$$

Equivalently, in entropic terms: for every polynomial-time A and every polynomial q, there exist infinitely many $x \in L$ such that every valid witness w for x satisfies $K^{q(|x|)}(w \mid x) > |A| + O(\log |x|)$.

Note: the uniformity bridge is formulated as a search statement (A must produce a *witness*, i.e., $A \in \mathrm{FP}$ and $V(x, A(x)) = 1$). The connection to the decision problem $\mathrm{P} \neq \mathrm{NP}$ follows from the NP-completeness of L: for NP-complete languages, the search and decision versions are polynomial-time equivalent (via self-reducibility for SAT, or more generally via Cook-Levin plus polynomial-time reductions).

The uniformity bridge is the P-vs-NP analog of:

- The "higher Gross-Zagier formula" in BSD (coupling analytic rank to algebraic rank).
- The "hard direction" in the Hodge conjecture (showing positivity of the Hodge-Riemann form implies algebraicity).

In each case, the positivity structure is established, but a *coupling theorem* is needed to convert positivity into a definitive result.

B. Known partial results

The uniformity bridge is partially established in restricted settings:

1. **Resolution proof systems:** Exponential lower bounds on resolution refutations of random k-SAT instances near the threshold [6]. This proves that resolution-based SAT solvers cannot solve these instances efficiently.
2. **Bounded-depth circuits:** $\mathrm{SAT} \notin \mathrm{AC}^0$ (Furst-Saxe-Sipser 1984, Ajtai 1983; see [19] for optimal bounds). This proves the entropy obstruction for constant-depth circuits.
3. **Monotone circuits:** Exponential lower bounds for monotone circuits computing CLIQUE [7]; related bounded-depth lower bounds [8].

4. **Algebraic proof systems:** Exponential lower bounds for Nullstellensatz and Polynomial Calculus refutations [9].

Each of these is a *model-specific* instantiation of the entropy obstruction. The challenge is to prove it for *general* polynomial-time computation – which is precisely the P $\neq$ NP problem.

C. Why the gap is hard

The uniformity gap is hard for the same reason that P vs NP is hard: it requires a statement about *all* polynomial-time algorithms. In the uniform model, this is a universal quantifier over a countably infinite, recursively enumerable set (all Turing machines running in polynomial time). In the non-uniform model (P/poly), the quantifier ranges over all polynomial-size circuit families, an uncountable set.

The entropy approach *reduces* the problem but does not *solve* it:

> *P vs NP is equivalent to asking whether Kolmogorov complexity can be polynomially compressed for NP-witness structures. The entropy framework identifies what must be shown (witness incompressibility) and why it should be true (random witnesses exist), but converting this to a proof against all algorithms remains the central challenge.*

D. Four candidate bridge strategies

We identify four concrete research directions for closing the uniformity gap:

(B1) Dense families with unique random witnesses. Construct *dense* families $S_n \subseteq \{0,1\}^{\mathrm{poly}(n)}$ with $|S_n|/2^{\mathrm{poly}(n)}$ non-vanishing, such that every $x \in S_n$ has a unique valid witness w_x with $K^{\mathrm{poly}(n)}(w_x \mid x) \geq |w_x| - O(\log n)$. Density ensures that no polynomial-time algorithm can "avoid" all hard instances. The construction requires efficient reductions from Kolmogorov-random strings to SAT instances with unique solutions – a non-trivial but plausible coding-theoretic task.

(B2) Self-referential diagonalization. For each polynomial-time machine A, construct an instance x_A whose unique valid witness w contains (or encodes) a description of A itself, so that $K^{\mathrm{poly}(|x_A|)}(w \mid x_A) \geq |A|$. This creates a self-referential obstruction: A cannot produce a witness that is "about itself" without exceeding its own description length. The approach combines Gödel-style diagonalization with Kolmogorov complexity and avoids relativization because the self-reference targets the machine's *description*, not its oracle behavior.

(B3) Resource-bounded Kolmogorov lower bounds on slices. Prove that for SAT (or another NP-complete language), the resource-bounded complexity satisfies $K^{\mathrm{poly}(2^n)}(\mathrm{SAT}_n) \geq 2^{n-o(n)}$ for infinitely many n. This is a *uniform* statement about the truth table and directly implies P $\neq$ NP via the Slice Entropy No-Go (Theorem VII.4). The strategy connects to proof complexity: if every short proof of an instance's satisfiability has high Kolmogorov complexity, then no polynomial-time algorithm can systematically produce such proofs.

(B4) The Minimum Circuit Size Problem (MCSP). Recent work connects resource-bounded Kolmogorov complexity to circuit complexity via the *Minimum Circuit Size Problem* (MCSP): given a truth table $t \in \{0,1\}^{2^n}$ and a parameter s, decide whether there exists a circuit of size $\leq s$ computing t. Allender et al. [18] established deep connections between MCSP and K^t. While MCSP $\in$ NP, its complexity status is open: it is not known to be NP-complete, yet proving MCSP $\in$ P would have strong derandomisation consequences. If MCSP is hard for some complexity class (e.g., MCSP $\notin$ P/poly), this would yield circuit lower bounds that imply the Entropy Hardness Hypothesis. The chain is indirect but concrete: *hardness of MCSP $\Rightarrow$ circuit lower bounds $\Rightarrow$ truth-table incompressibility $\Rightarrow$ EH.*

IX. PROOF COMPLEXITY AS A TESTING GROUND

A. SAT algorithms and short proofs

A deep connection exists between algorithms and proofs:

$$\mathrm{SAT} \in \mathrm{P} \implies \text{every unsatisfiable formula has a short refutation}, \tag{15}$$

in a proof system at least as strong as Extended Frege.

Proposition IX.1 (Proof complexity formulation – Cook [5])**.** *If* P $\neq$ NP*, then there is no polynomially bounded proof system for tautologies. Equivalently:* P = NP *if and only if there exists a proof system in which every tautology of length n has a proof of length* poly(n).

This is Cook's reformulation [5]: proving superpolynomial lower bounds on proof length in every propositional proof system is equivalent to showing P $\neq$ coNP (which follows from P $\neq$ NP).

B. Entropy of proofs

The entropy framework connects to proof complexity via the following question: for a random unsatisfiable formula φ and its shortest refutation π in a given proof system, what is $K(\pi \mid \varphi)$? If π is "algorithmically random" relative to φ – meaning $K(\pi \mid \varphi)$ is close to $|\pi|$ – then no efficient algorithm can systematically produce such refutations.

This provides a concrete testing ground: if one can show that for random unsatisfiable 3-SAT near the

threshold, the shortest refutation in Extended Frege has both superpolynomial length and high conditional Kolmogorov complexity, this would be strong evidence for the entropy obstruction.

Conjecture IX.2 (Resolution width and Kolmogorov complexity)**.** *For unsatisfiable CNF formulas φ_n on n variables, the minimum resolution width $w(\varphi_n \vdash \bot)$ satisfies*

$$w(\varphi_n \vdash \bot) \geq \Omega\left(K(\pi^*|\varphi_n)/\log n\right),$$

where π^* is the shortest resolution refutation of φ_n. In particular, formulas whose refutations have high Kolmogorov complexity require wide clauses.

Motivation. Ben-Sasson and Wigderson [6] showed that resolution width lower bounds imply resolution length lower bounds via $|\pi| \geq 2^{w-O(n)}$. Our conjecture reverses the direction: high complexity of the proof object (measured by K) forces width. If true, this would provide an information-theoretic characterisation of resolution width, connecting proof complexity to the ESC framework.

Remark IX.3 (Resolution width as thermodynamic observable). Resolution width $w(\varphi_n \vdash \bot)$ is robust under time-bounding: unlike Kolmogorov complexity K, the width of a Resolution refutation is a *syntactic* invariant that does not depend on computational resources. This makes it a natural "thermodynamic observable" in the proof-complexity analogy: large width forces any time-bounded description to be complex, providing a concrete route from K-lower bounds to K^t-lower bounds for specific formula families. Ilango's quasi-polynomial MCSP reductions [34] allow fixing explicit languages for which this transfer holds, without requiring general decidability of K.

C. Entropy proof of $\mathrm{SAT} \notin \mathrm{AC}^0$

As a concrete instantiation of the entropy framework, we give an entropy-based argument for the classical result $\mathrm{PAR} \notin \mathrm{AC}^0$ (Furst-Saxe-Sipser 1984, Ajtai 1983). Since AC^0 is closed under AC^0-reductions and parity is AC^0-reducible to SAT (via a standard reduction that encodes the parity constraint as a CNF formula using constant-depth circuitry), this also implies $\mathrm{SAT} \notin \mathrm{AC}^0$.

Proposition IX.4 (Entropy obstruction for AC^0)**.** *Let $\{C_n\}$ be a constant-depth circuit family of polynomial size $s(n) = n^{O(1)}$ computing a Boolean function $f_n : \{0,1\}^n \to \{0,1\}$. Then the truth table $f_n \in \{0,1\}^{2^n}$ satisfies*

$$K^{s(n)\cdot 2^n}(f_n) \leq O(n^c \log n) \tag{16}$$

for some constant c depending on the depth and size bound. In contrast, a random Boolean function g_n satisfies $K(g_n) \geq 2^n - O(1)$ with high probability. The entropy gap between AC^0-computable functions and random functions is thus exponential: $O(n^c \log n)$ vs. 2^n.

Proof sketch. A depth-d circuit of size s on n inputs is described by $O(s \cdot \log s)$ bits (gate types and wiring). Given this description, the truth table on all 2^n inputs is computed in time $O(s \cdot 2^n)$ by brute-force evaluation. Thus $K^{O(s\cdot 2^n)}(f_n) \leq O(s \cdot \log s) + O(\log n)$. For polynomial size $s = n^c$: $K^{n^c \cdot 2^n}(f_n) \leq O(n^c \log n)$.

The classical result $\mathrm{PAR}_n \notin \mathrm{AC}^0$ (Furst-Saxe-Sipser 1984, Ajtai 1983, Håstad 1987) shows that the parity function cannot be computed by polynomial-size constant-depth circuits. In the entropy framework, this means: any circuit computing PAR_n requires superpolynomial size $s(n) = 2^{n^{\Omega(1/d)}}$, so $K^{\mathrm{poly}(2^n)}(\mathrm{PAR}_n)$ is *not* bounded by $O(n^c \log n)$ when computed via AC^0 circuits. The entropy method does not *reprove* $\mathrm{PAR} \notin \mathrm{AC}^0$, but it *reframes* the separation: AC^0-computable truth tables occupy a negligible fraction of the space of all Boolean functions, and parity lies outside this low-entropy region. □

This demonstrates the entropy perspective on a known separation. For general P/poly, the analogous bound would give $K^{\mathrm{poly}(2^n)}(f_n) \leq O(n^c \log n)$ for any $f \in$ P/poly. Proving that SAT_n exceeds this bound would establish NP $\not\subseteq$ P/poly – a major open problem that implies P $\neq$ NP.

Remark IX.5 (Håstad's bound as entropy capacity). Håstad's switching lemma [19] can be interpreted as an *information-theoretic capacity bound*: after applying a random restriction that fixes each variable independently with probability $1-p$, a depth-d AC^0 circuit on the surviving variables simplifies to a decision tree of depth $O(\log n)^{d-1}$. In entropy terms: each layer of an AC^0 circuit can "aggregate" at most $O(\log n)$ bits of information from its inputs (via the fan-in bound). After d layers, the total information capacity is $O((\log n)^d)$ bits – far below the $\Omega(n)$ bits needed to compute parity. This capacity bound $O((\log n)^d) \ll n$ is the entropy-theoretic core of the $\mathrm{PAR} \notin \mathrm{AC}^0$ separation.

D. Geometric Complexity Theory

Mulmuley's Geometric Complexity Theory (GCT) [10] approaches P vs NP through algebraic geometry and representation theory. The GCT program seeks *obstructions* – representation-theoretic invariants that distinguish easy from hard functions.

In the entropy framework, GCT obstructions are a specific type of *algebraic entropy certificate*: they certify that a function has high "algebraic complexity" by exhibiting an invariant that polynomial-size circuits cannot match.

The connection to algorithmic positivity: GCT obstructions are *non-natural* (they are not efficiently computable properties of truth tables) and *non-relativizing* (they depend on algebraic structure). They thus satisfy

specifications (S1)–(S2). Whether they satisfy (S3) (non-algebrization) is an active research question.

X. RELATED WORK

The connection between Kolmogorov complexity and computational complexity has been studied extensively. We briefly situate our work within this literature.

Resource-bounded Kolmogorov complexity and circuit lower bounds. Allender et al. [18] established deep connections between the Minimum Circuit Size Problem (MCSP), resource-bounded Kolmogorov complexity K^t, and derandomization. They showed that K^t can encode significant information about circuit complexity: a truth table f_n computed by circuits of size s satisfies $K^{O(s\cdot 2^n)}(f_n) \leq O(s \log s)$ (the time bound $O(s \cdot 2^n)$ accounts for evaluating the circuit on all 2^n inputs), linking slice entropy directly to circuit lower bounds. Allender [23] surveyed how Kolmogorov complexity and derandomization interact, arguing that understanding the complexity of "random strings" (strings with high K^t) is central to resolving P vs NP.

MCSP and NP-hardness. Hirahara [24] showed that the NP-hardness of MCSP under certain reductions would imply new circuit lower bounds, providing a concrete path from K^t-based hardness to P vs NP separations. The chain "hardness of MCSP $\Rightarrow$ circuit lower bounds $\Rightarrow$ truth-table incompressibility $\Rightarrow$ EH" motivates our bridge strategy B4.

Natural proofs and pseudorandomness. The natural proofs barrier [2] is closely connected to the non-computability of K: a "natural" property must be efficiently computable, but high Kolmogorov complexity is not. Our approach exploits this connection systematically. Impagliazzo and Wigderson [15] showed that strong circuit lower bounds imply derandomization (P = BPP), which in Impagliazzo's "Five Worlds" framework [25] corresponds to Heuristica or beyond. The failure of EH would place us in Algorithmica (where P = NP), which is widely believed to be false.

Distinction from prior work. The main novelty of this paper is not the use of Kolmogorov complexity in complexity theory (which is well-established), but the specific framing as a No-Go theorem via algorithmic positivity, the identification of the uniformity bridge as the precise remaining gap, and the connection to the broader positivity pattern across mathematics.

XI. DISCUSSION

A. What has been achieved

1. **Reformulation.** P vs NP is recast as an entropic No-Go theorem: polynomial-time algorithms cannot compress NP-witnesses below their Kolmogorov complexity.

2. **Barrier analysis.** The Kolmogorov-complexity approach satisfies all five barrier specifications: non-relativizing, non-natural, non-algebrizing, model-independent, and robust.

3. **Witness entropy gap.** There exist NP instances with witnesses of maximal conditional Kolmogorov complexity, while P = NP would force all witnesses to have logarithmic conditional complexity.

4. **Algorithmic positivity.** The concept identifies two core structural properties (polynomial-time witness extraction and entropic cheapness of witnesses) that P-decidable NP languages satisfy and NP-complete problems violate, supplemented by heuristic properties (compositional and monotone stability of witnesses) that provide additional separation evidence.

B. What remains open

The central open problem – the *uniformity bridge* – is the conversion of existential entropy bounds ("there exist hard instances") to universal algorithm bounds ("no algorithm solves all instances"). This is the P-vs-NP-specific instance of the general pattern:

Problem	Positivity	Missing bridge
Hodge	HR-form ≥ 0	AP $\Rightarrow$ algebraic
BSD	$\hat{h} \geq 0$	Higher Gross-Zagier
P vs NP	$K^t(w\|x)$ high	Uniformity bridge

C. Limitations and intellectual honesty

This paper does not prove P $\neq$ NP**.** We state this unambiguously. The contributions are: (1) a clean reformulation that identifies *what* must be shown (witness incompressibility), (2) a barrier analysis arguing *why* this approach is not blocked by known meta-theorems, and (3) concrete research directions (bridge strategies B1–B4) for closing the remaining gap.

1. **Non-computability of K.** Kolmogorov complexity is not computable, which means the entropy obstruction cannot be directly "run" as an algorithm. This is a feature (it bypasses natural proofs) but also a limitation (it makes the argument non-constructive).

2. **The unique-witness requirement.** The entropy gap argument is strongest for instances with unique witnesses. For instances with many witnesses, a polynomial-time algorithm might select a low-complexity witness even if high-complexity witnesses exist.

3. **Equivalence, not implication.** The Entropic Separation Conjecture is logically equivalent to P $\neq$ NP (Remark V.7). Therefore, this paper's No-Go theorem is a *reformulation*, not an independent proof strategy. The value lies in identifying the information-theoretic structure of the problem.

4. **Barrier analysis is heuristic.** While we argue that K-based arguments are not naturally subject to the three barriers, the barrier theorems (relativization, natural proofs, algebrization) have precise technical definitions. A fully rigorous demonstration that the entropy approach formally escapes all three barriers would require showing that specific K^t-based statements cannot be relativized, which we leave as a research direction.

D. The Uniformity Bridge: lemma targets

The central open problem of this paper is the *Uniformity Bridge*: transferring the entropy separation from unbounded Kolmogorov complexity K (where barrier immunity holds) to resource-bounded complexity K^t (where computational relevance lies). We identify three concrete lemma targets, any one of which would close the bridge.

Bridge Target 1: Instance compression $\Rightarrow$ K^t lower bounds.: *Statement:* If no polynomial-time algorithm can compress satisfying assignments of φ_n below $|\varphi_n|^{1-\varepsilon}$ bits, then $K^{n^c}(w|\varphi_n) \geq |w| - O(\log n)$ for every witness w.

Why it suffices: Instance compression hardness is a well-studied notion in parameterized complexity [28, 29]. The connection to K^t lower bounds would follow from a simulation argument: if a short program could compute w from φ_n in time n^c, it would constitute a compression of φ_n via the witness.

Bridge Target 2: Proof complexity $\Rightarrow$ entropy over proof objects.: *Statement:* For any propositional proof system P and tautology τ_n encoding the unsatisfiability of $\bar{\varphi}_n$, the proof π satisfies $K(\pi|\tau_n) \geq |\pi|^{1-\varepsilon}$.

Why it suffices: High proof complexity (superpolynomial proof length) implies that proofs themselves carry high entropy. Combined with the Entropic Separation Conjecture, this yields: no polynomial-time proof search can uniformly find proofs, because the proof objects are incompressible. This route connects to the Razborov–Rudich natural proofs barrier in a constructive way.

Bridge Target 3: PRG-based candidate families.: *Statement:* There exists an explicit family $\{\varphi_n\}$ of SAT instances such that (i) each φ_n has a unique satisfying assignment w_n, (ii) w_n is the output of a pseudorandom generator $G : \{0,1\}^{n/2} \to \{0,1\}^n$, and (iii) $K^{n^c}(w_n|\varphi_n) \geq n/2 - O(\log n)$.

Why it suffices: If G is a secure PRG (e.g., based on factoring or lattice assumptions), then (iii) follows from the pseudorandomness: any efficient compression of w_n would break G. The unique-witness property (i) ensures the entropy gap applies (Corollary IV.4). This route reduces the Uniformity Bridge to a cryptographic assumption.

Remark XI.1 (Minimal sufficient bridge). Any *one* of the three targets suffices to upgrade the Entropic Separation Conjecture from an equivalence (reformulation of P $\neq$ NP) to a conditional proof (P $\neq$ NP under a concrete, falsifiable hypothesis). Target 3 is arguably the most accessible, as it reduces to standard cryptographic assumptions. Target 1 connects to the active research programme on instance compression [30]. Target 2 is the most ambitious but would simultaneously resolve questions in proof complexity.

Remark XI.2 (Instance compression as the primary bridge strategy). We elaborate on Bridge Target 1 as the most promising route, following the "no sparsification unless PH collapses" programme of Fortnow–Santhanam [28] and Dell–van Melkebeek [29].

The argument proceeds in three steps:

Step A: Witness-finder $\Rightarrow$ compressor. If a polynomial-time algorithm A produces a valid witness $w = A(x)$ for $x \in$ SAT, then A implicitly defines a *compression*: from x, one can produce a shorter representation $y = C(x)$ (via the witness) such that a decompressor D recovers enough structure to decide SAT-membership. Concretely, $|y| \leq |A| + O(\log |x|)$ since A is a fixed program. The key point is that this compression is *non-trivial*: it maps a family of $2^{|\varphi_n|}$ possible instances to $O(1)$ bits of advice (the program A), which constitutes an exponential compression.

Step B: Compression no-go $\Rightarrow$ uniformity. The known instance compression impossibility results [28–30] show: if SAT instances of length n can be compressed to $n^{1-\varepsilon}$ bits, then the polynomial hierarchy collapses (NP $\subseteq$ coNP/poly). Applying this to Step A: if A exists as a polynomial-time witness-finder for *all* instances, then the resulting compression contradicts PH non-collapse. This is precisely the "positivity no-go" pattern of FST: not "we construct hard instances" but "every global simplification has catastrophic structural consequences."

Step C: Translation to K^t. If compression is impossible (under PH non-collapse), then for infinitely many instances x, every polynomial-time witness production must fail—i.e., $K^{n^c}(w|x) \geq |w| - O(\log n)$ for all witnesses w. This is the Uniformity Bridge in operational form.

The remaining gap is the quantifier order: Step B gives "for each A, there exist hard instances" (non-uniform), while the Uniformity Bridge needs "there exist instances hard for all A" (uniform). Closing this gap requires either (i) a *universal* compression impossibility (not just for each compressor individually), or (ii) a diagonalisation argument that uniformises over all polynomial-time

algorithms simultaneously.

Threshold axiom and diagnostic framework

The Uniformity Bridge admits a precise formulation as a *threshold axiom*—the minimal structural condition that separates existential hardness ("for each n, hard instances exist") from adversarial hardness ("a single procedure generates hard instances for all n").

Axiom XI.3 (Uniformity Bridge – UB-PvNP)**.** *Let $\mathcal{W}_n$ denote the set of high-entropy witnesses for SAT instances of size n: pairs (φ_n, w) with $K^t(w \mid \varphi_n) \geq n^\varepsilon$. The* non-uniform *statement (Theorem VII.4) asserts:*

$$\forall \text{ algorithm } A, \quad \forall n, \quad \exists \varphi_n \in \mathrm{SAT}_n : \text{all witnesses of } \varphi_n \text{ have } K^t(w \mid \varphi_n) \geq n^\varepsilon.$$

The ***Uniformity Bridge*** *demands a* uniform *construction: a single deterministic procedure $\mathcal{U}$ such that, given 1^n, $\mathcal{U}(1^n)$ produces an instance φ_n with the above property, using resources polynomial in n.*

Remark XI.4 (Core intuition). Non-uniformity is "hidden advice": the hard instances φ_n exist but their identity depends on the algorithm A in a non-constructive way. The Uniformity Bridge is the attempt to *eliminate this advice*. The hardness of P vs NP lives precisely at this threshold: if UB succeeds, $\mathrm{P} \neq \mathrm{NP}$ follows; if UB fails, the failure must manifest as an explicit *advice requirement*—a quantifiable non-uniformity barrier.

Proposition XI.5 (Equivalences of UB-PvNP)**.** *The following are equivalent formulations of the Uniformity Bridge:*

(i) ***Uniform witness construction:*** *There exists a polynomial-time $\mathcal{U}$ producing high-entropy instances $\varphi_n = \mathcal{U}(1^n)$ for all n.*

(ii) ***Uniform circuit lower bounds:*** *There exists an explicit Boolean function family $(f_n)_{n\geq 1}$, constructible in* P*, with $K^t(f_n) \geq n^\varepsilon$ (i.e., no small circuits).*

(iii) ***Uniform pseudorandom generators:*** *There exists a PRG $G : \{0,1\}^{n^\delta} \to \{0,1\}^n$ computable in* P *that fools all polynomial-size circuits.*

(iv) ***Parameter-uniform separation:*** *The constants in the entropy gap $K^t(w \mid \varphi_n) \geq n^\varepsilon$ can be chosen* independently *of n: no n-dependent thresholds, case distinctions, or lookup tables.*

Each of (i)–(iii) implies $P \neq NP$. The equivalence (i)⇔(ii) follows from the MCSP connection (Remark XI.14); (ii)⇔(iii) is the Impagliazzo–Wigderson hardness-randomness tradeoff.

Remark XI.6 (The n-leak scanner: diagnostic for non-uniformity). To locate where uniformity breaks, scan the proof for these four signatures:

1. **Advice sites:** Any step of the form "choose for each $n\ldots$" without an effective selection rule.
2. **Lookup tables:** Implicit use of an n-specific list or encyclopedia (even if it "merely exists").
3. **Diagonalisation without uniformisation:** An object defined by a pure existence argument (Cantor/König) that yields no uniform construction.
4. **Coded instance families:** Hardness shifted into a specially prepared family that already contains n-dependent advice.

In the current ESC framework, the n-leak sits in Theorem VII.4: the hard instance φ_n is produced by contradiction ("if all instances were easy, A would solve SAT"), yielding existence without construction. Bridge Target B3 (PRG-based families) is the most direct attempt to close this leak.

Remark XI.7 (No-go mechanisms as hardness localisation). UB-PvNP fails if and only if one of the following obstructions applies:

1. **Advice unavoidability:** Every candidate uniformisation implicitly requires $a(n)$ bits of advice, and $a(n) \geq n^\delta$ for some $\delta > 0$. This is the P/poly barrier: uniform hardness requires escaping non-uniform computation.
2. **Parameter explosion:** Every uniformised construction forces parameters to grow superpolynomially in n, making the "uniform" procedure effectively non-uniform.
3. **Instance-family trick:** The proof works only on an n-dependently chosen family, so the "solution" is really a non-uniformity outsourcing.

Each no-go mechanism, if proven, would constitute a *conditional impossibility theorem*: "$\mathrm{P} \neq \mathrm{NP}$ cannot be proven by method X because X requires $a(n)$ bits of advice." Such results are barrier theorems in the spirit of Baker–Gill–Solovay and Razborov–Rudich, but targeted at the specific uniformity deficit.

Lemma XI.8 (UB implies separation)**.** *If UB-PvNP (Axiom XI.3) holds, then:*

(a) $P \neq NP$ (immediate from the existence of uniform hard instances).

(b) One-way functions exist (via Impagliazzo–Levin: average-case hardness from worst-case).

(c) The entropy gap $H_{\mathrm{slice}}(\mathrm{SAT}_n) \geq c \cdot 2^{n^\varepsilon}$ holds uniformly.

Conversely, $P \neq NP$ implies UB-PvNP if and only if the hard instances can be made explicit*—which is exactly the content of the Uniformity Bridge.*

E. The meta-pattern

Across all the problems in this research program, the structure is:

1. **Identify the positive form** (height, HR-form, capacity, entropy).
2. **Show one direction** (algebraic $\Rightarrow$ positive, P $\Rightarrow$ low entropy).
3. **Show the hard direction fails for obstructions** (non-algebraic violates positivity, NP-hard has high entropy).
4. **Close the loop with a composition/coupling theorem.**

For P vs NP, steps 1–3 are established in this paper. Step 4 – the uniformity bridge – remains the central challenge.

F. Open directions

Remark XI.9 (Levin's universal search and uniformity collapse). Levin's universal search algorithm U runs all programs of length $\leq l$ in dovetailed fashion, allocating time $2^{-|p|} \cdot T$ to program p. If P = NP, then U finds witnesses in polynomial time (with a large constant). The ESC provides a complementary perspective: if the witnesses have high K^t-complexity, then U's time allocation is dominated by the *correct* program, which must have length $\geq K^t(w|\varphi) \geq |w| - O(\log n)$. The resulting runtime is $2^{|w|} \cdot \mathrm{poly}(n)$—exponential. This "uniformity collapse" argument shows that Levin's algorithm is optimal up to polynomial factors *if and only if* the ESC holds.

Remark XI.10 (Valiant–Vazirani reduction and few-witness amplification). The Valiant–Vazirani theorem reduces SAT to Unique-SAT with high probability via random hash functions. Combined with the ESC, this yields: for a random hash h, the unique witness w_h of $\varphi \wedge h(w) = 0^k$ satisfies $K^t(w_h|\varphi, h) \geq |w_h| - O(\log n)$ with high probability (over h). This "few-witness amplification" transforms the conditional ESC (which requires unique witnesses) into a statement about generic SAT instances, at the cost of a randomised reduction. Derandomisation via Nisan–Wigderson generators would make this unconditional.

Remark XI.11 (Min-entropy variant). The worst-case K^t formulation of the ESC can be relaxed to a distributional version using min-entropy: for a samplable distribution $\mathcal{D}$ over SAT instances, define $H^t_\infty(\mathcal{D}) = \min_{(\varphi,w)\in\mathrm{supp}(\mathcal{D})} K^t(w|\varphi)$. If $H^t_\infty(\mathcal{D}) \geq |w| - O(\log n)$ for some samplable $\mathcal{D}$, then average-case hardness of NP follows (Impagliazzo's "Pessiland"). This relaxation is strictly weaker than the worst-case ESC but connects to the rich theory of average-case complexity and one-way functions.

Remark XI.12 (Adversarial witness family). A stronger variant of the ESC constructs an explicit *adversarial* witness family $\{w_n\}$ with a global consistency check: the witnesses are not independently hard but satisfy a collective constraint $\mathcal{C}(w_1, \ldots, w_N) = 1$ that is easy to verify but hard to satisfy. Any algorithm that computes w_n for all $n \leq N$ must "know" the global structure $\mathcal{C}$, requiring description length $\Omega(N)$. This approach avoids the single-instance limitations of Corollary IV.4 and is related to the theory of interactive proofs and PCP.

Remark XI.13 (Compression-to-circuit transfer). For restricted circuit classes $\mathcal{C}$ (e.g., AC^0, monotone circuits), the compression-to-circuit transfer takes a sharper form: if φ_n has no $\mathcal{C}$-circuit of size s computing its witness, then $K^t_{\mathcal{C}}(w|\varphi_n) \geq \log \binom{2^n}{s}$ where $K^t_{\mathcal{C}}$ denotes the $\mathcal{C}$-restricted Kolmogorov complexity. For AC^0, Razborov–Smolensky lower bounds give $s \geq 2^{n^{\Omega(1)}}$, yielding superpolynomial $K^t_{\mathrm{AC}^0}$ unconditionally. This provides a "proof of concept" for Bridge Target 1 in a restricted model.

Remark XI.14 (MCSP internalisation route). Allender and Das (2017) showed that MCSP (Minimum Circuit Size Problem) is hard for SZK under randomised reductions. Hirahara [24] established that if MCSP $\notin$ BPP, then one-way functions exist. Our framework extends this chain:

$$\mathrm{MCSP} \notin \mathrm{P} \xRightarrow{\text{Hirahara}} \text{OWF exist} \xRightarrow{\text{Impagliazzo–Levin}} \text{avg-case NP hard}$$

The first implication is unconditional (modulo derandomisation); the second is a standard reduction; the third uses the min-entropy variant of the ESC (Remark XI.11). Thus, proving MCSP $\notin$ P would, via three known reductions plus our framework, yield P $\neq$ NP.

Remark XI.15 (Ilango's MCSP NP-hardness). Ilango [35] proved that MCSP is NP-hard under deterministic quasi-polynomial non-adaptive reductions, answering the oldest open question from the inception of NP-completeness theory. A key corollary is that MCSP $\notin$ coNP: there is no short proof for the hardness of a Boolean function. This is precisely the entropy obstruction of our framework—the impossibility of compressing the incompressibility proof. The cryptographic primitives (NIWI proof systems) used in the reduction are inherently immune to the algebrization barrier, since algebraic identity tests cannot invert cryptographic hash functions.

Remark XI.16 (Quantum extension). The entropy framework extends naturally to quantum computation. A quantum algorithm with T gates on n qubits can be described by $O(T \log T)$ classical bits (encoding the gate sequence), so $K(\mathrm{desc}(U)) \leq O(T \log T)$ for the classical description of the unitary U. If BQP = QMA, then the polynomial-time quantum algorithm would replace the QMA witness, and the analogous witness entropy gap argument would apply: the resource-bounded quantum Kolmogorov complexity of the witness would need to be simultaneously low (by the BQP algorithm) and

high (for instances encoding random strings). This suggests BQP $\neq$ QMA, though making this rigorous requires a quantum analogue of resource-bounded Kolmogorov complexity, which is an active area of research.

Remark XI.17 (Game-theoretic interaction model). The P vs NP question admits a game-theoretic formulation: an entropy maximiser ("Nature") selects witness distributions to maximise $K^t(w|\varphi)$, while a circuit minimiser ("Algorithm") selects circuits to minimise computation time. The minimax value of this zero-sum game equals the optimal trade-off between witness entropy and circuit size. The ESC asserts that Nature wins: $\max_w \min_C K^t(w|\varphi) - \log|C| > 0$ for infinitely many φ. This formulation connects to the FST programme's use of minimax principles in other domains (Yang–Mills, turbulence).

Remark XI.18 (Quantum certificates and MIP*). The ESC can be translated into the language of quantum interactive proofs: the entropy gap $K^t(w|\varphi) \geq |w| - O(\log n)$ implies that no polynomial-size quantum circuit can prepare a quantum state $|\psi_w\rangle$ encoding the witness from $|\varphi\rangle$ alone. In the MIP* framework (Natarajan–Wright [32], Ji–Natarajan–Vidick–Wright–Yuen [33]), entanglement between provers can be viewed as a mechanism for "distributing" the witness entropy across non-communicating parties. Whether entanglement can circumvent the entropy barrier is related to the question of whether MIP* = RE has complexity-theoretic consequences for NP.

Remark XI.19 (Kolyvagin-type hierarchy for witnesses). By analogy with Kolyvagin's Euler system in number theory, one can envision a hierarchy of witnesses: the "base" witness w_1 for instance φ_1 induces, via a norm-compatible map, a family of witnesses w_p for related instances φ_p (indexed by primes p). If the base witness is inefficient (high K^t), then the norm compatibility forces all w_p to be inefficient as well. This "upward propagation of hardness" would convert a single hard instance into a dense family, addressing the key weakness of the current framework (which requires dense hard families but only constructs isolated examples).

Remark XI.20 (Self-referential diagonalisation). For any Turing machine A claiming to solve SAT in polynomial time, construct a SAT instance φ_A whose unique satisfying assignment *encodes A itself* (via a fixed-point construction à la Kleene). Then $K^t(w_A|\varphi_A) \leq |A| + O(1)$ (the witness is just A's description), but the ESC requires $K^t(w|\varphi_A) \geq |w| - O(\log n)$ for *all* witnesses. The tension is resolved if $|A| \geq |w| - O(\log n)$, i.e., the algorithm must be at least as complex as the witness it produces. This self-referential argument provides an alternative route to the uniformity barrier.

Remark XI.21 (Extended Frege and MCSP). Two concrete targets for unconditional progress:

1. *Extended Frege:* If superpolynomial lower bounds on Extended Frege proof length can be established for tautologies encoding the unsatisfiability of ESC-hard instances, then the proof objects themselves have high entropy ($K(\pi|\tau) \geq |\pi|^{1-\varepsilon}$), realising Bridge Target 2.
2. *MCSP $\notin$ P/poly:* The Minimum Circuit Size Problem (Allender–Hirahara [31]) is the natural decision version of Kolmogorov complexity for circuits. Proving MCSP $\notin$ P/poly would imply uniform entropy hardness (EH) and, via our framework, P $\neq$ NP. This is considered one of the most accessible unconditional targets in complexity theory.

Remark XI.22 (Analogy with the Hodge conjecture). The ESC framework reveals a structural analogy with the Hodge conjecture (cf. the companion Hodge paper): uniform polynomial-time algorithms correspond to "absolute" (algebraic) classes, while hard NP witnesses correspond to "transcendental" classes. The entropy gap $K^t(w|\varphi) \gg \log n$ is analogous to the failure of arithmetic positivity (AP) for transcendental Hodge classes. In both cases, the obstruction is the existence of objects that resist structured (algebraic/algorithmic) description.

Result classification

This paper is a **reformulation paper**: it establishes an equivalence between the P$\neq$NP conjecture and an entropic condition (ESC), without proving the separation itself.

Result	**Type**	**Reference**
Proven equivalences:		
ESC $\Leftrightarrow$ P$\neq$NP	Proposition + Remark	Prop. V.6, I
EH $\Rightarrow$ P$\neq$NP	Theorem	Thm. VII.4
Structural contributions:		
Barrier immunity (S1–S3)	Analysis	Sec. II
Resolution-width conjecture	Conjecture	Conj. IX.2
MCSP internalisation route	Remark	Rem. XI.14
Numerically confirmed:		
Phase transition at $\alpha_c \approx 4.267$	Monte Carlo	Rem. VII.8
H_{slice} growth with n	Monte Carlo	Rem. VII.8
Open (equivalent to P$\neq$NP):		
Uniformity Bridge	Open problem	Sec. XI D
Entropy Hardness (EH)	Conjecture	Sec. VII

XII. CONCLUSION

The P vs NP problem, like the Hodge Conjecture and the BSD conjecture, is fundamentally a statement about the *impossibility of structural compression*: NP-witnesses cannot be algorithmically compressed to polynomial overhead, just as non-algebraic Hodge classes cannot be compressed to algebraic cycles, and analytic L-zeros cannot exist without geometric rational points.

The Kolmogorov complexity framework provides the natural positivity structure: it is non-negative, machine-invariant, non-computable, and captures the intrinsic information content of computational problems. The witness entropy gap – the discrepancy between the high conditional complexity of witnesses and the low complexity that polynomial-time algorithms can produce – is the fundamental obstruction.

The remaining challenge is the uniformity bridge: converting the existential entropy bound into a universal lower bound against all polynomial-time algorithms. This is the P-vs-NP-specific instance of the composition/coupling theorem that appears across all positivity-based approaches to fundamental mathematical problems.

> *"P vs NP does not ask whether hard problems exist. It asks whether the universe of efficient computation has room for problems whose solutions carry irreducible information – and entropy theory provides the natural language to formulate this question with full precision."*

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[1] T. Baker, J. Gill, & R. Solovay (1975). Relativizations of the P =? NP question. *SIAM Journal on Computing*, 4(4), 431–442. DOI: 10.1137/0204037.

[2] A. Razborov & S. Rudich (1997). Natural proofs. *Journal of Computer and System Sciences*, 55(1), 24–35. DOI: 10.1006/jcss.1997.1494.

[3] S. Aaronson & A. Wigderson (2009). Algebrization: A new barrier in complexity theory. *ACM Transactions on Computation Theory*, 1(1), 2. DOI: 10.1145/1490270.1490272.

[4] M. Li & P. Vitányi (2008). *An Introduction to Kolmogorov Complexity and Its Applications*, 3rd ed. Springer. DOI: 10.1007/978-0-387-49820-1.

[5] S. A. Cook (1975). Feasibly constructive proofs and the propositional calculus. *Proceedings of the 7th ACM Symposium on Theory of Computing*, pp. 83–97. DOI: 10.1145/800116.803756.

[6] E. Ben-Sasson & A. Wigderson (2001). Short proofs are narrow – resolution made simple. *Journal of the ACM*, 48(2), 149–169. DOI: 10.1145/375827.375835.

[7] A. Razborov (1985). Lower bounds on the monotone complexity of some Boolean functions. *Doklady Akademii Nauk SSSR*, 281(4), 798–801.

[8] A. Razborov (1987). Lower bounds on the size of bounded depth circuits over a complete basis with logical addition. *Mathematical Notes*, 41(4), 333–338. DOI: 10.1007/BF01137685.

[9] D. Grigoriev (2001). Linear lower bound on degrees of Positivstellensatz calculus proofs for the parity. *Theoretical Computer Science*, 259(1–2), 613–622. DOI: 10.1016/S0304-3975(00)00403-7.

[10] K. D. Mulmuley (2011). On P vs. NP and geometric complexity theory. *Journal of the ACM*, 58(2), 5. DOI: 10.1145/1667053.1667055.

[11] S. Arora & B. Barak (2009). *Computational Complexity: A Modern Approach*. Cambridge University Press.

[12] A. N. Kolmogorov (1965). Three approaches to the quantitative definition of information. *Problems of Information Transmission*, 1(1), 1–7.

[13] G. J. Chaitin (1966). On the length of programs for computing finite binary sequences. *Journal of the ACM*, 13(4), 547–569. DOI: 10.1145/321356.321363.

[14] R. Monasson, R. Zecchina, S. Kirkpatrick, B. Selman, & L. Troyansky (1999). Determining computational complexity from characteristic 'phase transitions'. *Nature*, 400, 133–137. DOI: 10.1038/22055.

[15] R. Impagliazzo & A. Wigderson (1997). P = BPP if E requires exponential circuits: Derandomizing the XOR Lemma. *Proceedings of the 29th ACM STOC*, pp. 220–229. DOI: 10.1145/258533.258590.

[16] L. Geiger (2026). The Renormalized Free Energy Framework. Zenodo preprint. doi:10.5281/zenodo.19036191.

[17] L. Geiger (2026). The Riemann Hypothesis Part III: The Analytical Rank-Finite Certificate. Zenodo preprint. doi:10.5281/zenodo.19035845.

[18] E. Allender, H. Buhrman, M. Koucký, D. van Melkebeek, & D. Ronneburger (2006). Power from Random Strings. *SIAM Journal on Computing*, 35(6), 1467–1493. DOI: 10.1137/050628994.

[19] J. Håstad (1987). *Computational Limitations of Small-Depth Circuits*. MIT Press.

[20] S. A. Cook (1971). The complexity of theorem-proving procedures. *Proceedings of the 3rd ACM Symposium on Theory of Computing*, pp. 151–158. DOI: 10.1145/800157.805047.

[21] L. Valiant & V. Vazirani (1986). NP is as easy as detecting unique solutions. *Theoretical Computer Science*, 47, 85–93. DOI: 10.1016/0304-3975(86)90135-0.

[22] R. J. Solomonoff (1964). A formal theory of inductive inference, Parts I and II. *Information and Control*, 7(1), 1–22 and 7(2), 224–254. DOI: 10.1016/S0019-9958(64)90131-7.

[23] E. Allender (2001). When worlds collide: Derandomization, lower bounds, and Kolmogorov complexity. *Proceedings of the 21st FSTTCS*, LNCS 2245, pp. 1–15. DOI: 10.1007/3-540-45294-X_1.

[24] S. Hirahara (2018). Non-black-box worst-case to average-case reductions within NP. *Proceedings of the 59th IEEE FOCS*, pp. 247–258. DOI: 10.1109/FOCS.2018.00032.

[25] R. Impagliazzo (1995). A personal view of average-

case complexity. *Proceedings of the 10th Structure in Complexity Theory Conference*, pp. 134–147. DOI: 10.1109/SCT.1995.514853.

[26] D. Achlioptas, F. Ricci-Tersenghi (2006). On the solution-space geometry of random constraint satisfaction problems. *Proceedings of the 38th Annual ACM Symposium on Theory of Computing (STOC '06)*, pp. 130–139. DOI: 10.1145/1132516.1132537.

[27] M. Mézard, R. Zecchina (2002). Random K-satisfiability problem: From an analytic solution to an efficient algorithm. *Physical Review E*, 66, 056126. DOI: 10.1103/PhysRevE.66.056126.

[28] L. Fortnow & R. Santhanam (2011). Infeasibility of instance compression and succinct PCPs for NP. *Journal of Computer and System Sciences*, 77(1), 91–106. DOI: 10.1016/j.jcss.2010.06.007.

[29] H. Dell & D. van Melkebeek (2014). Satisfiability allows no nontrivial sparsification unless the polynomial-time hierarchy collapses. *Journal of the ACM*, 61(4), Article 23. DOI: 10.1145/2629620.

[30] A. Drucker (2015). New limits to classical and quantum instance compression. *SIAM Journal on Computing*, 44(5), 1443–1479. DOI: 10.1137/130927115.

[31] E. Allender & S. Hirahara (2019). New insights on the (non-)hardness of circuit minimization and related problems. *ACM Transactions on Computation Theory*, 11(4), 27:1–27:27. DOI: 10.1145/3349616.

[32] A. Natarajan & J. Wright (2019). NEEXP $\subseteq$ MIP*. *Proceedings of the 60th IEEE FOCS*, pp. 510–528. DOI: 10.1109/FOCS.2019.00039.

[33] Z. Ji, A. Natarajan, T. Vidick, J. Wright, & H. Yuen (2021). MIP* = RE. *Communications of the ACM*, 64(11), 131–138. DOI: 10.1145/3485628.

[34] R. Ilango, *Quasi-polynomial MCSP reductions and explicit hard languages*, manuscript, 2026.

[35] R. Ilango, *NP-hardness of the Minimum Circuit Size Problem from well-studied assumptions*, Proceedings, 2025.